

Technical Skills

- Programming Languages: **JavaScript**, TypeScript, Python, C/C++, C#, Java, Golang
- Web Technologies: **React.js**, HTML5, CSS, Node.js, Next.js
- Game Development: **Unity**, **Unreal Engine**, SDL
- Database & Tools: **MongoDB**, **SQL**, Git, VS Code, Visual Studio
- Security & Infra Tools: Scapy, GDB, Wireshark, Vagrant, Metasploit, Terraform, **AWS CLI**

Technical Experience

JavaScript Developer Co-op **Semios** **Jan 2022 – Aug 2022**

- Built and maintained **React** components following **Agile** methodologies, ensuring consistent delivery quality.
- Built reusable UI components using **Ant Design** library, implementing features like **data tables**, forms, and **interactive map interfaces**.
- Resolved front-end bugs and implemented UI/UX improvements through effective collaboration with **QA team**.
- Enhanced application functionality by implementing robust **form validation**, content sorting, and date/time management using **moment.js**.

Web Developer **Bluvec Technologies Inc.** **Sep 2021 – Dec 2021**

- Developed company's landing page using **React**, ensuring responsive design and brand consistency.
- Implemented automated testing suite using **Golang** to improve code reliability.
- Built automated image management system using **GoCV** for timestamp processing and file cleanup.

Technical Projects

APEX **Python, MCP** **Jan 2026 - May 2026**

- Built an autonomous AI-driven penetration testing framework evaluated on **42 HackTheBox machines**, achieving a **71% success rate** (30/42) across Linux and Windows targets.
- Built a multi-agent pipeline where recon (nmap, gobuster, ZAP) and execution agents dispatch LLM-generated bash scripts to Kali Linux via a custom MCP-to-HTTP REST API bridge.
- Designed an LLM-based **failure classifier** (FIXABLE / FUNDAMENTAL / SCRIPT_ERROR) with three independent loop-detection counters to prevent infinite retry cycles and auto-fail stalled tasks.
- Implemented dual **RAG pipelines** — offensive payload retrieval and historical failure retrieval — to guide exploit selection and avoid previously failed attack paths.

Music Stats Page **Next.js, React, Tailwind CSS, Redis** **June 2025**

- Integrated APIs from Last.fm, iTunes, and Spotify to fetch top tracks/artists and enrich album artwork and links
- Built **responsive** React components display listening history with fallback handling
- Implemented **Redis caching** with 1-hour TTL to reduce API calls and support manual/automatic cache refresh
- Optimized data fetching using Promise.all and normalized JSON structures from multiple APIs

TCP Network Attack Simulation **Python, Scapy, Vagrant, GDB** **April 2025**

- Simulated and evaluated **TCP-layer attacks** in a 3-node virtual network using Vagrant.
- Launched a **TCP SYN flood** to exhaust server backlog and observed mitigation via SYN cookies.
- Executed a **TCP RST injection** to forcibly terminate active Telnet/SSH sessions.
- Crafted spoofed packets to **hijack Telnet sessions**, inject shell commands, and validate impact via packet capture and file system changes.

ROP Exploitation **C, Python, GDB, ROPgadget** **Mar 2025**

- Developed and tested **ROP chains** against a custom x64 Linux binary with **NX protection enabled and ASLR disabled**.
- Used **GDB and ROPgadget** to locate gadgets and construct precise exploit payloads, including handling register constraints and gadget chaining.
- Wrote Python scripts to automate ROP payload generation and gadget chaining using offsets from libc.

PrediQL	Python, GraphQL, Ollama	Sep 2024 – Dec 2024
- Developed an AI-assisted fuzzing pipeline targeting GraphQL endpoint vulnerabilities such as SQL Injection, DoS, and batching attacks. - Designed Python scripts to automate LLaMA-based query generation via Ollama. - Converted LLaMA outputs into structured JSON payloads and used them to test GraphQL APIs, iteratively refining queries based on error responses.		
Cloud Infrastructure Deployment	EC2, VPC, IAM, DynamoDB	Nov 2024
- Deployed a secure VPC network on AWS using the web console, including public/private subnets, Internet Gateway, NAT Gateway, Route Tables, and Security Groups. - Launched EC2 instances and configured a Node.js web server accessible on port 8081. - Created and managed IAM users, groups, and roles, applying the principle of least privilege. - Built a DynamoDB table and securely connected it to EC2 via role-based permissions. - Implemented Elastic IPs and studied their behavior on instance restarts.		
Serverless API Development	AWS Lambda	Nov 2024
- Built a serverless REST API using AWS Lambda and API Gateway, eliminating the need for EC2 servers. - Developed a Lambda function to support both GET and POST requests for a comment system. - Secured the POST endpoint via AWS Cognito, implementing JWT-based authorization. - Validated API with Postman, tested user-based access control, and discussed JWT tamper-resistance mechanisms.		
Eggly Buddy	C++, OpenGL, ECS Framework	Sep 2023 – Dec 2023
- Programmed interactive elements using C++, enabling in-game pet mechanics and mini-games. - Implemented visual effects using custom shaders, including dynamic parallax background and animations. - Developed save/load system using nlohmann/json library for persistent game state.		
Swag the North	Unity, C#	Sep 2023
Top 12 Finalist out of 200+ projects in Hack the North. - Spearheaded development of an immersive game using Unity and C#, featuring innovative swag collection mechanics. - Implemented core gameplay systems including player controls, item spawning, and scene management. - Led version control workflow and mentored team members in Git best practices.		
Locative Audio	Unity, C#	Jan 2023 – Apr 2023
- Developed a GPS-based mobile app that lets users discover Vancouver’s cultural audio content at specific locations. - Built an efficient content management system using XML data structure and integrated AR Foundation for image detection. - Created an interactive map interface with iGUI featuring custom camera controls and smooth user navigation.		
FREN	React, JavaScript, MongoDB, Express, Redux	Jun 2021 – Aug 2021
- Built a social networking app with user recommendation system and real-time chat functionality. - Implemented Redux state management and routing for seamless user experience. - Developed responsive chat interface with custom pop-up components for friend recommendations.		
Education		
Vancouver, BC	Simon Fraser University	Sep 2024 – May 2026
Master’s in Professional Computer Science (Focus: Cybersecurity)		
Vancouver, BC	University of British Columbia	Sep 2018 – May 2024
Graduated with a BS in Computer Science, with Distinction		